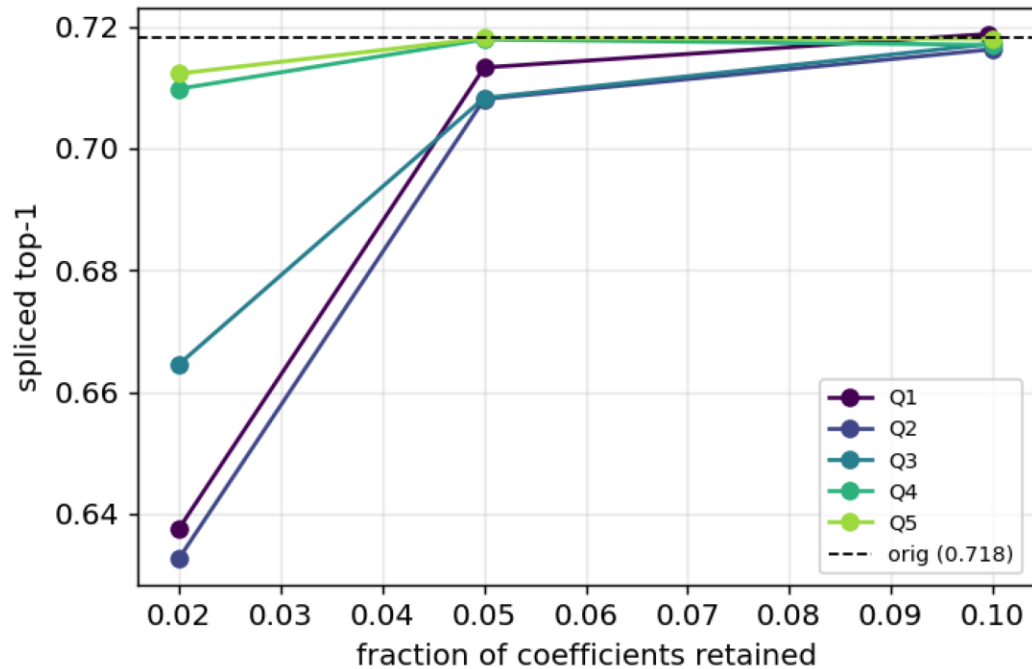
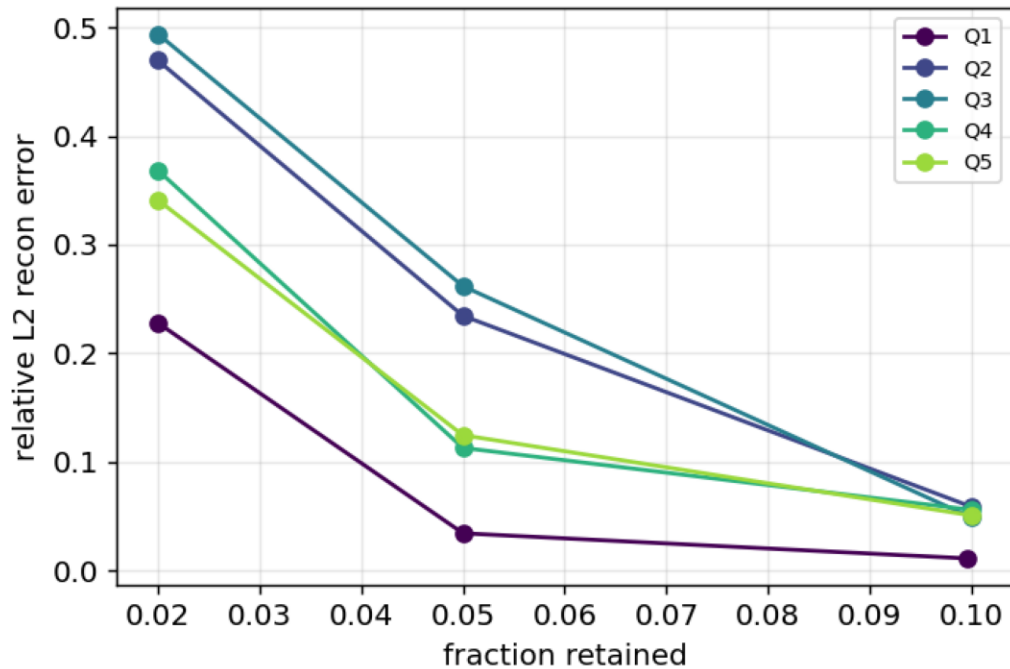


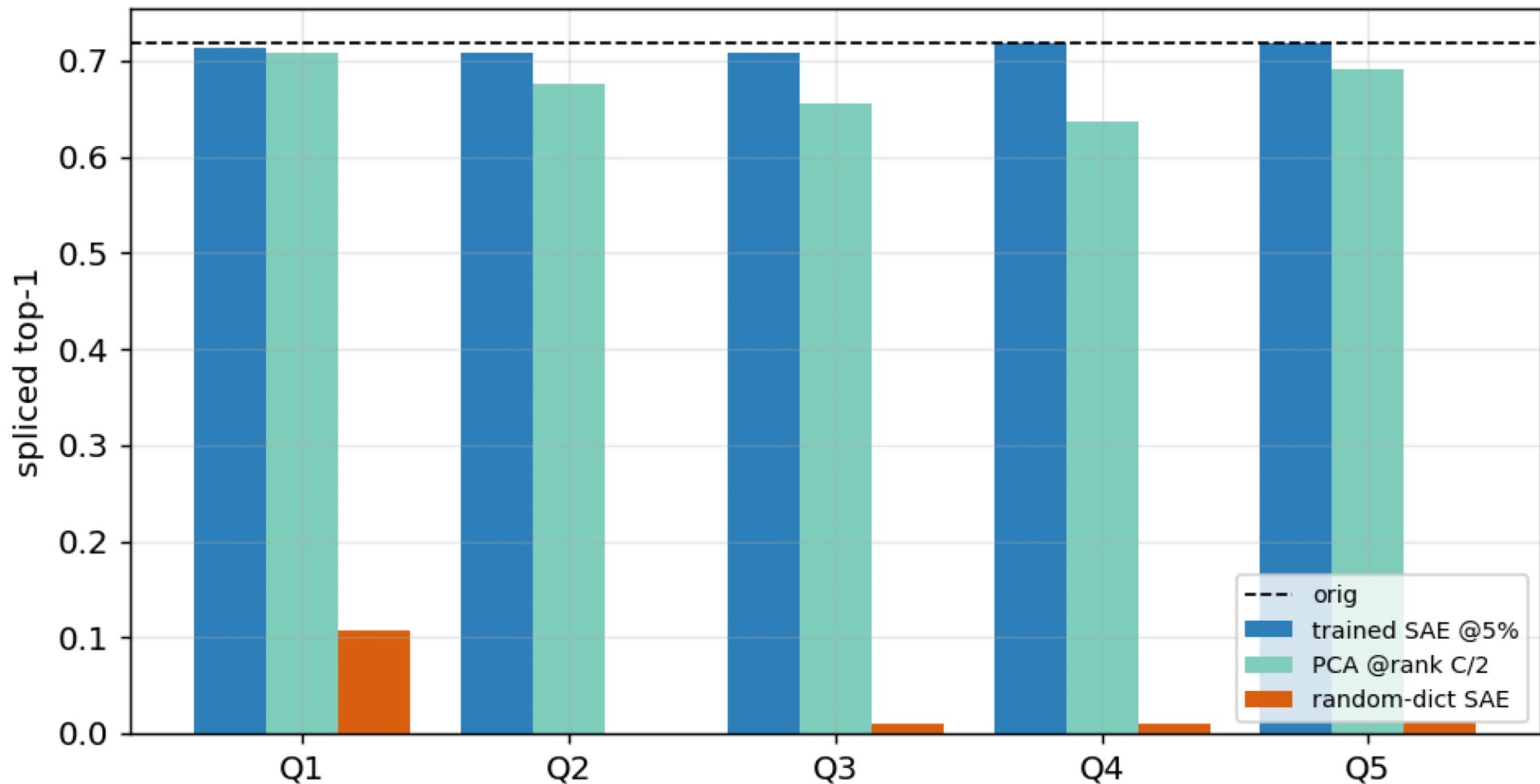
Phase 1: sparse reconstruction preserves accuracy
(Vector SAE + global mask)



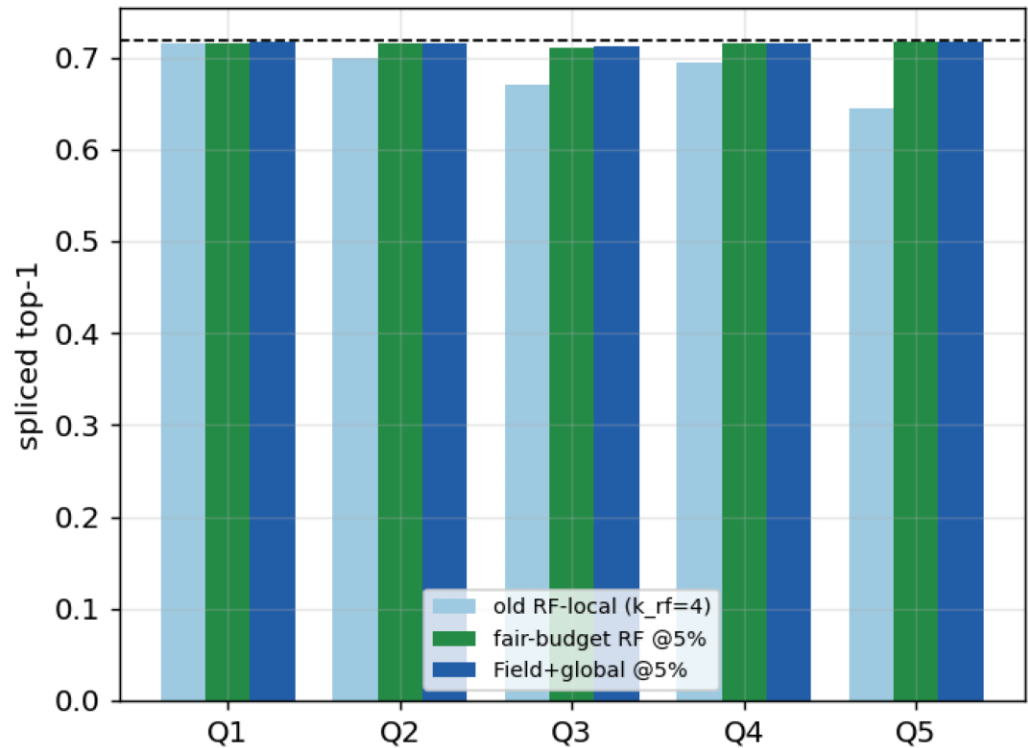
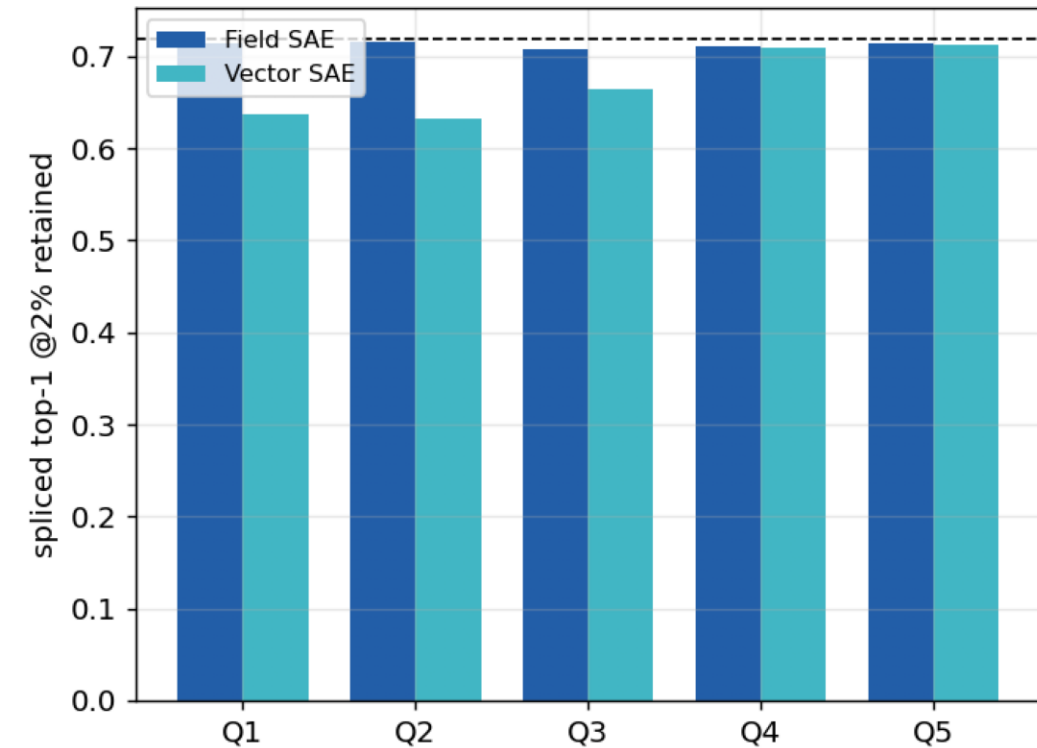
Reconstruction error vs sparsity



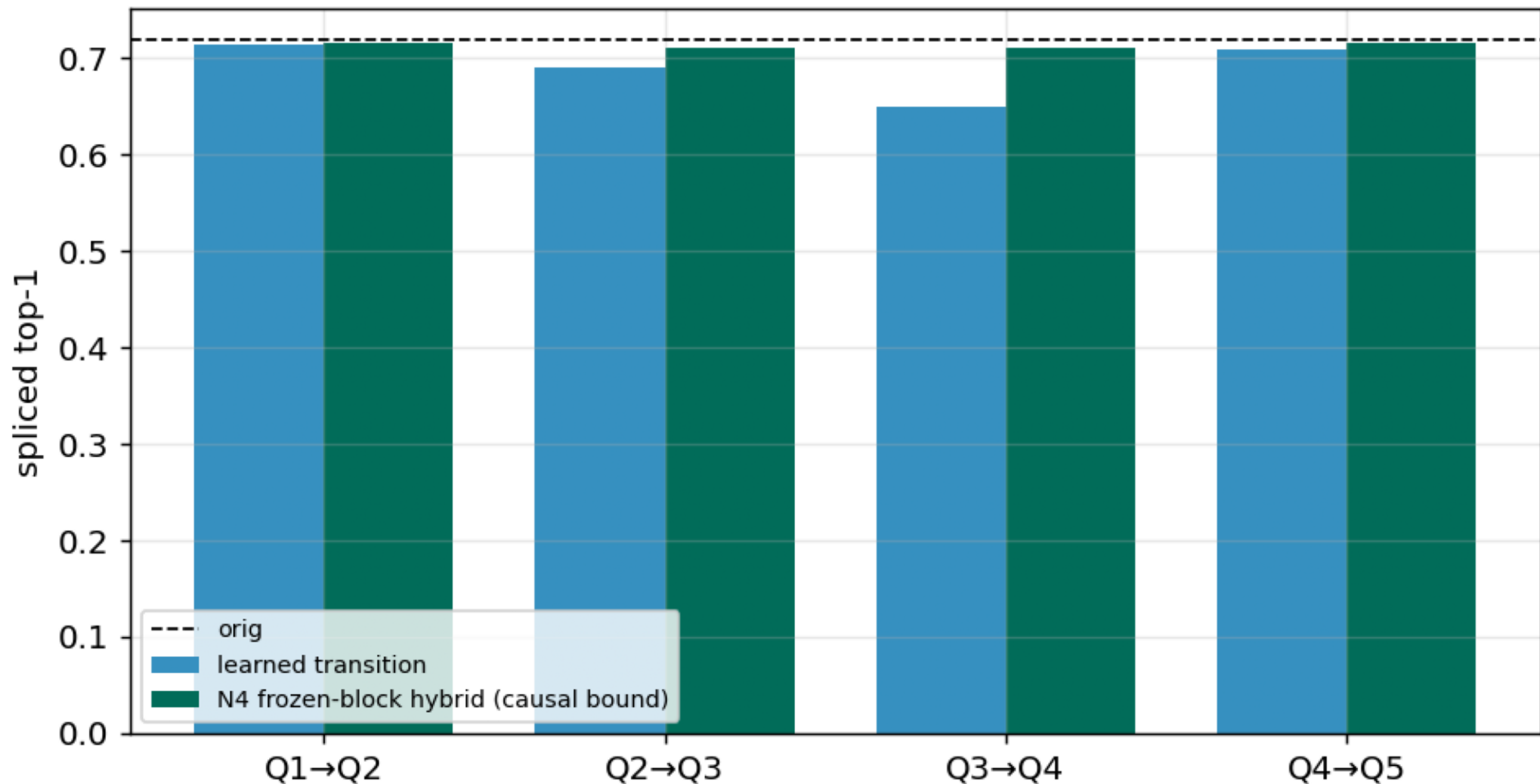
Phase 1: trained SAE >> PCA and random-dict baselines



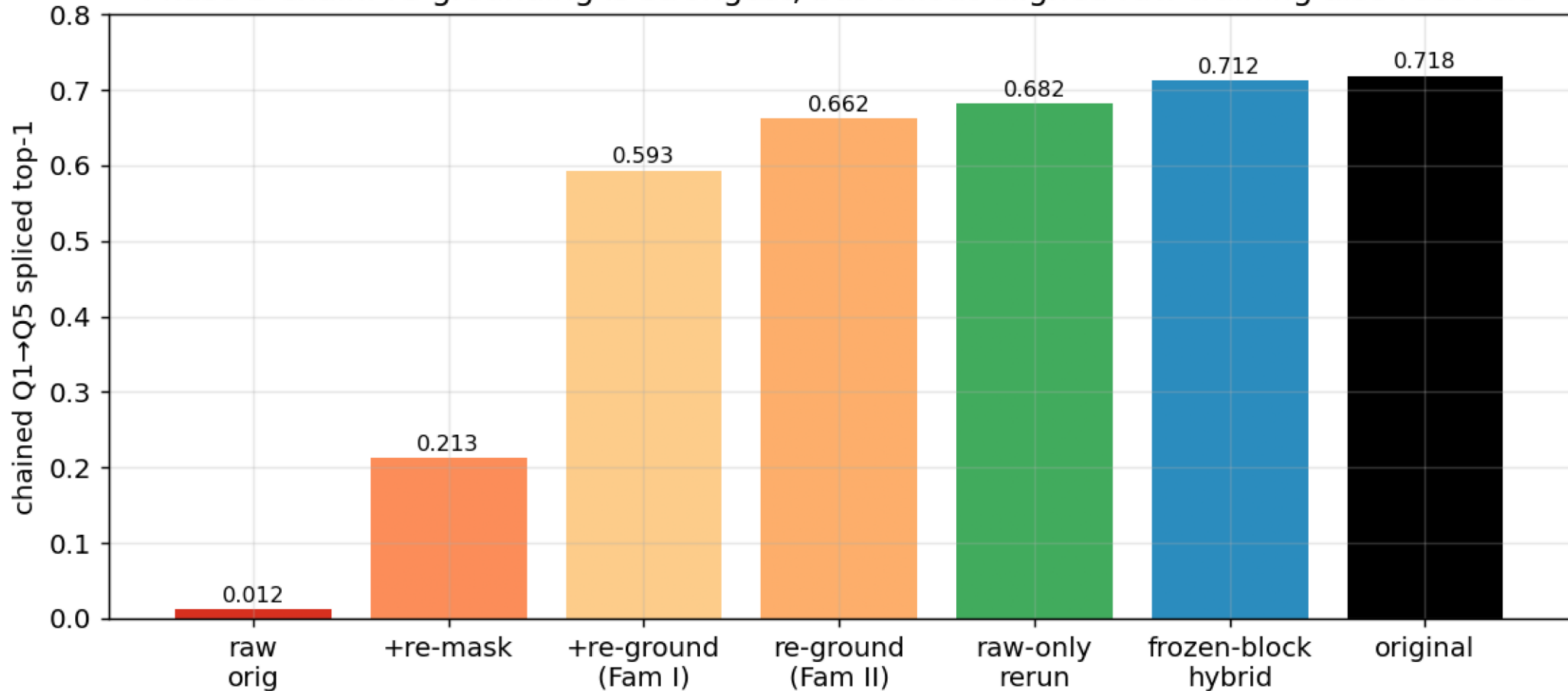
Field SAE still beats Vector at aggressive global sparsity. By local masking improves sharply once the budget is matched



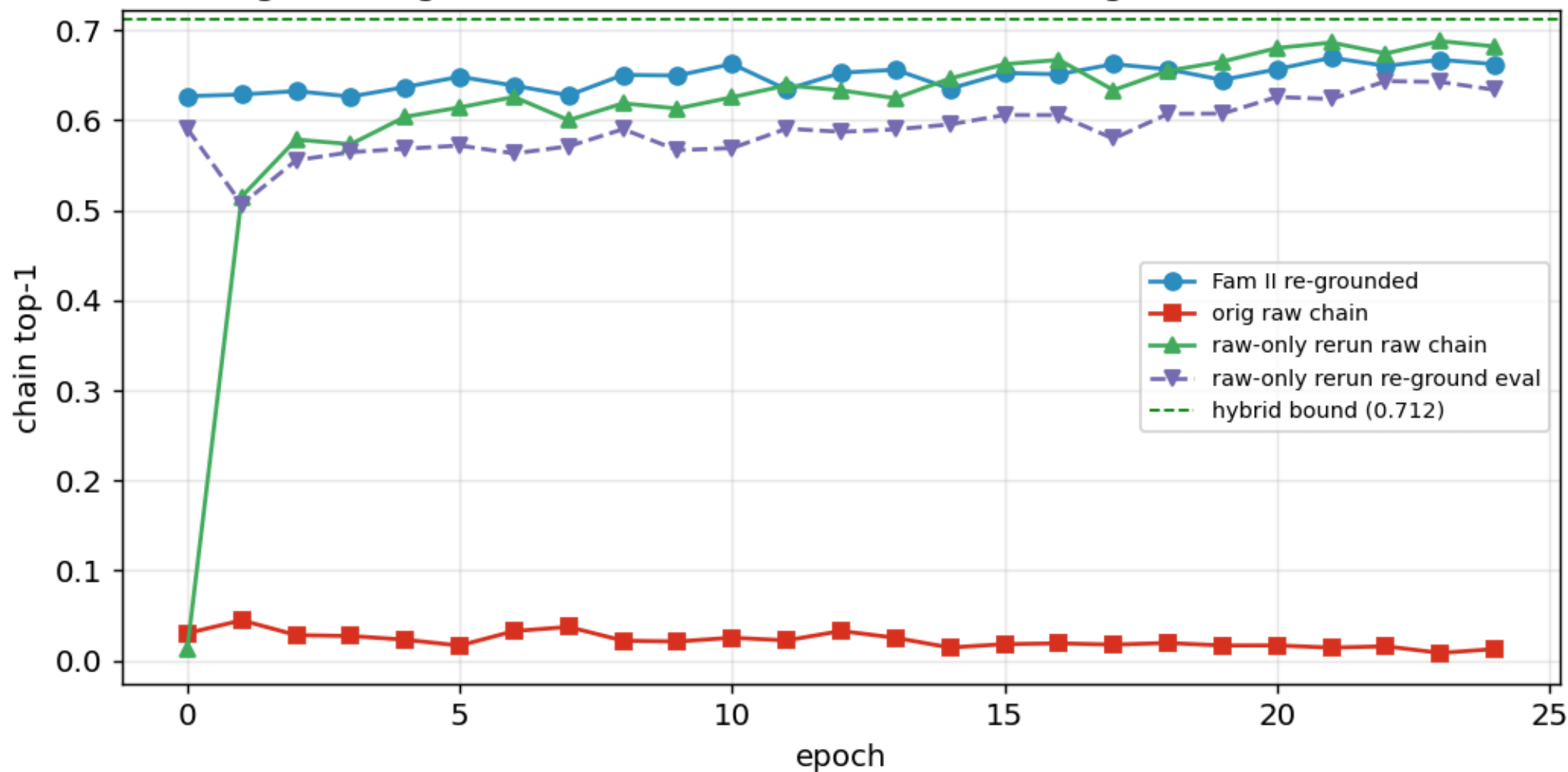
Phase 3: single-step transitions (learned vs causal upper bound)



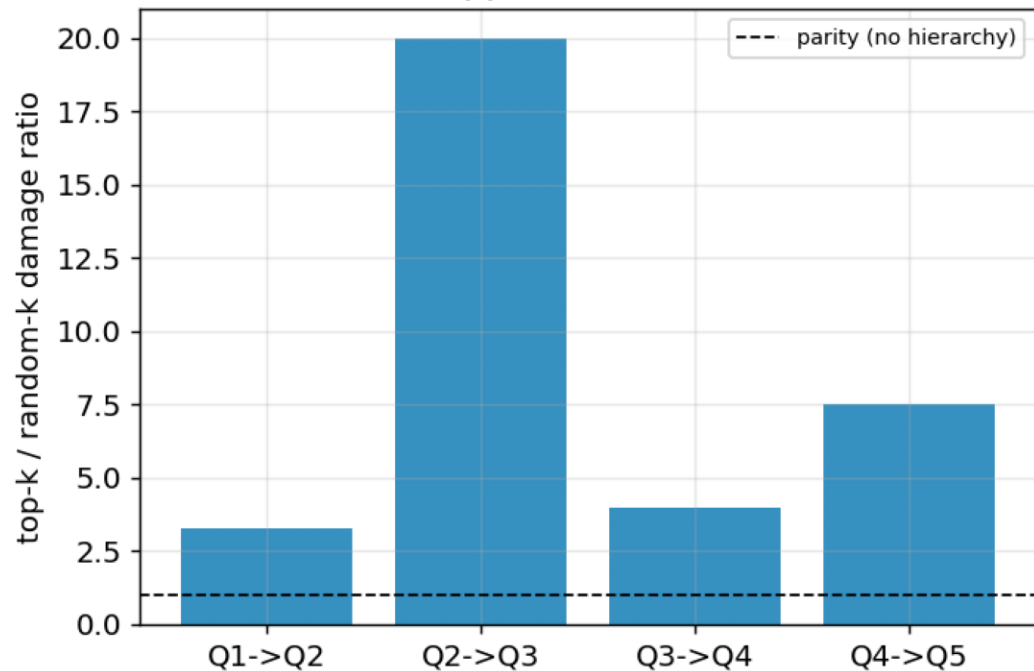
Phase 3 chain: re-grounding is strongest, but rollout-aligned raw training also recovers



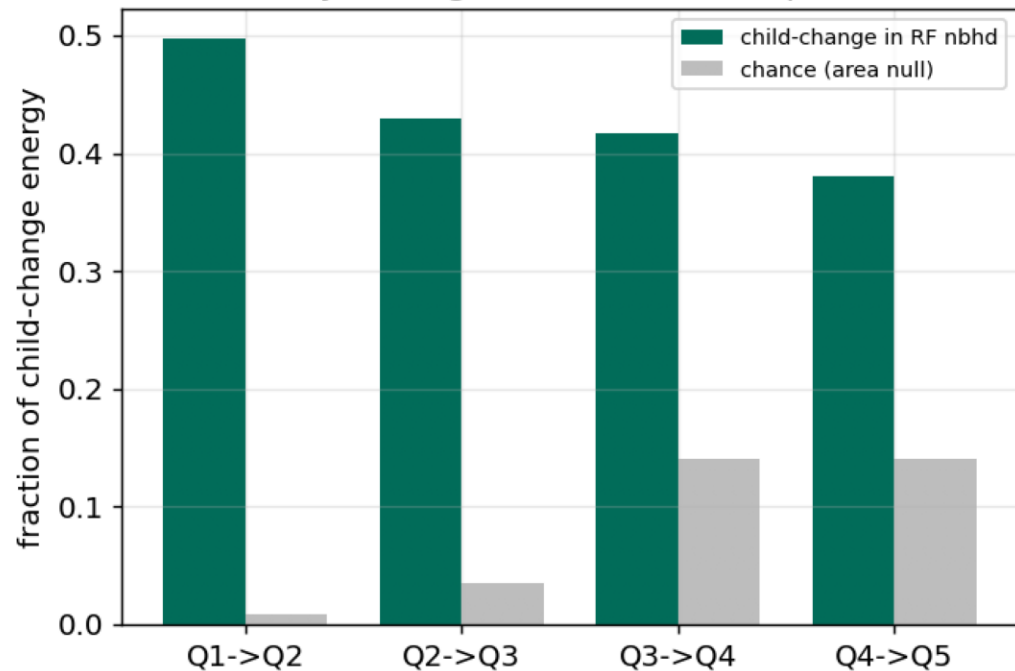
Phase 3b: re-grounding remains best, but raw rollout training removes the chance colla



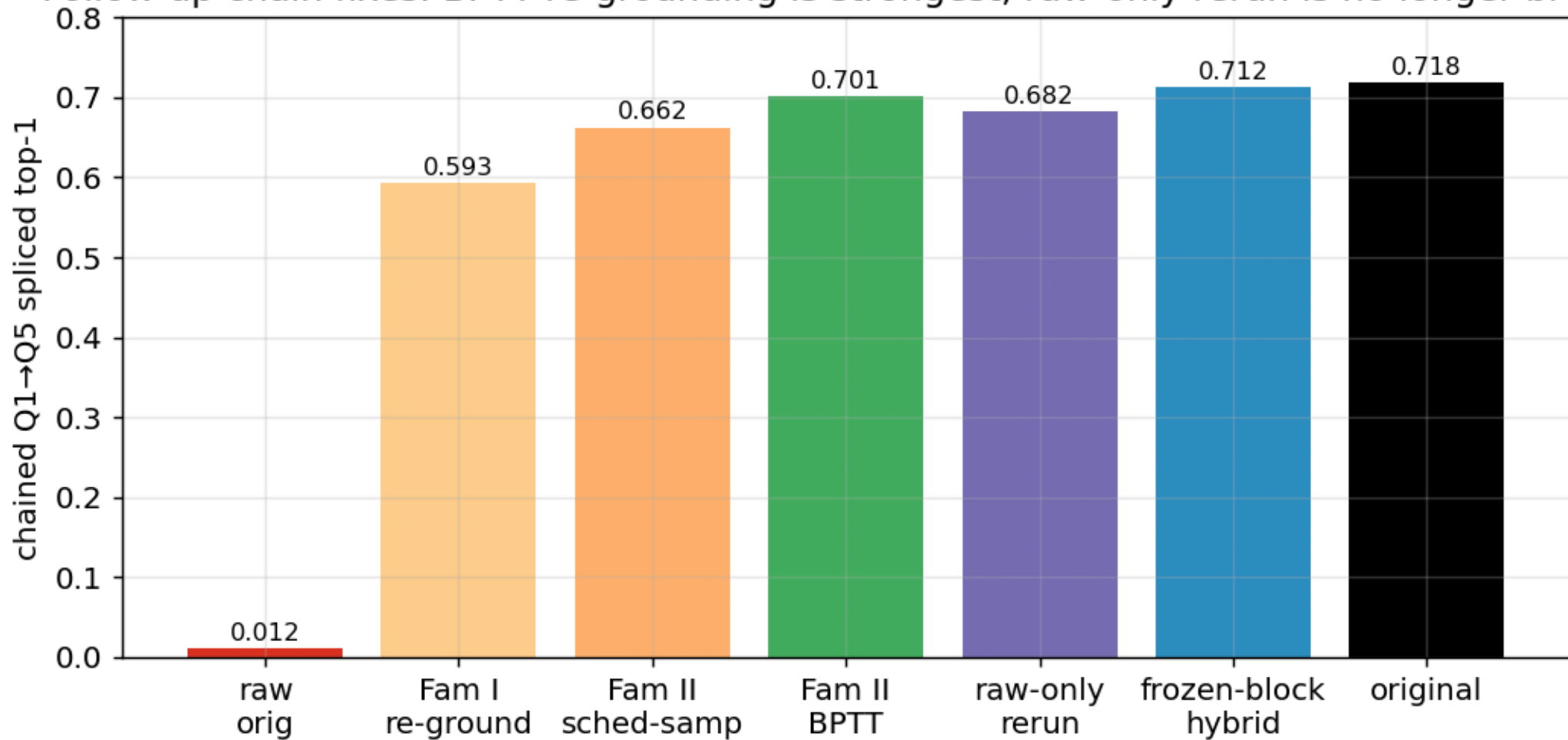
N8 importance: top parents >> random kept coeffs
(ratio clipped at 20; some $\rightarrow \infty$)



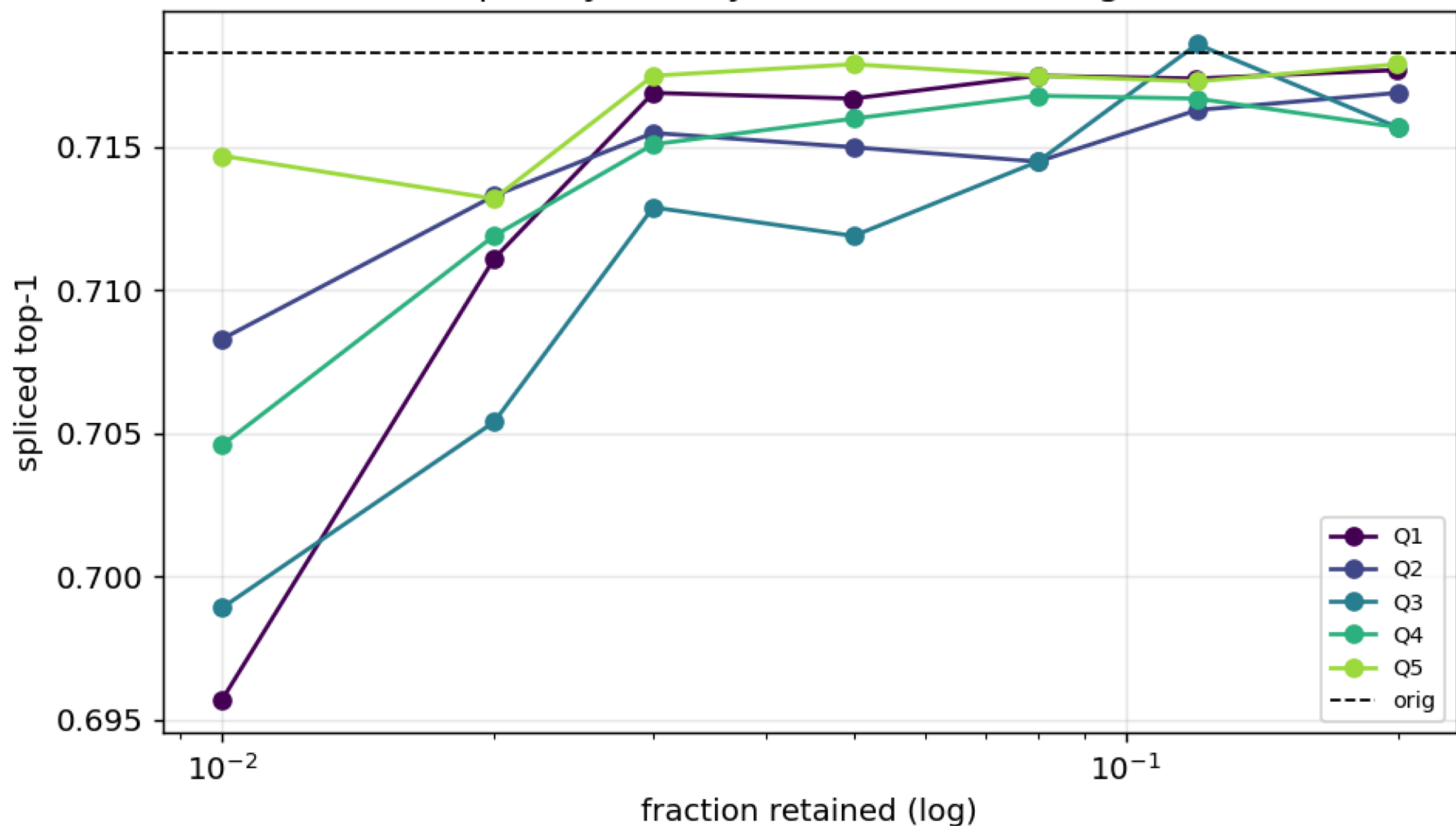
N8 locality: change concentrates in parent's RF



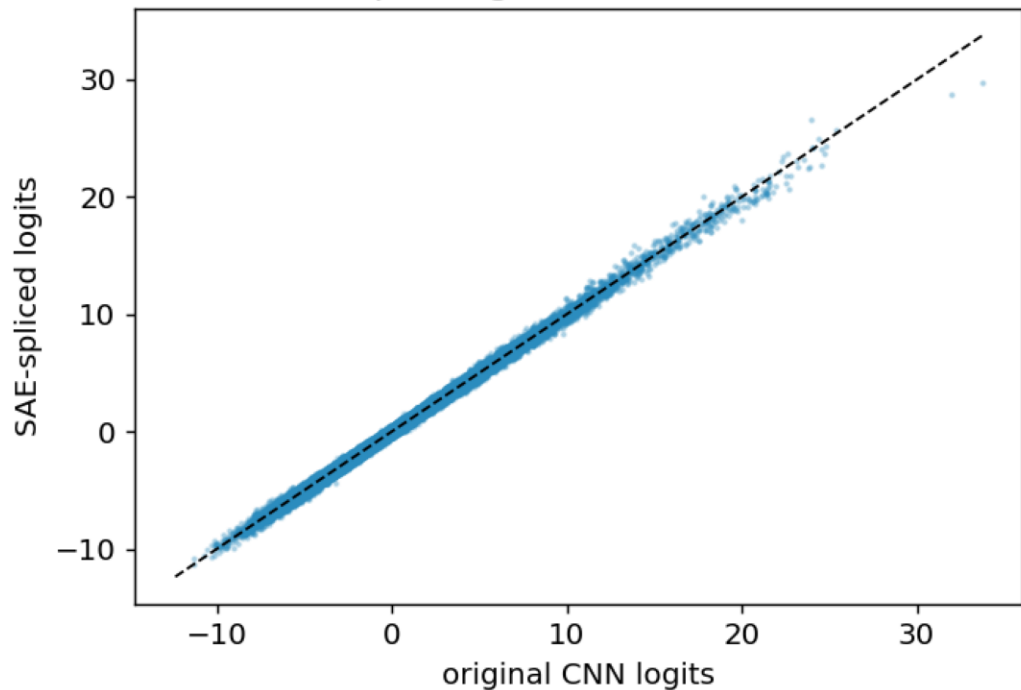
Follow-up chain fixes: BPTT re-grounding is strongest, raw-only rerun is no longer broken



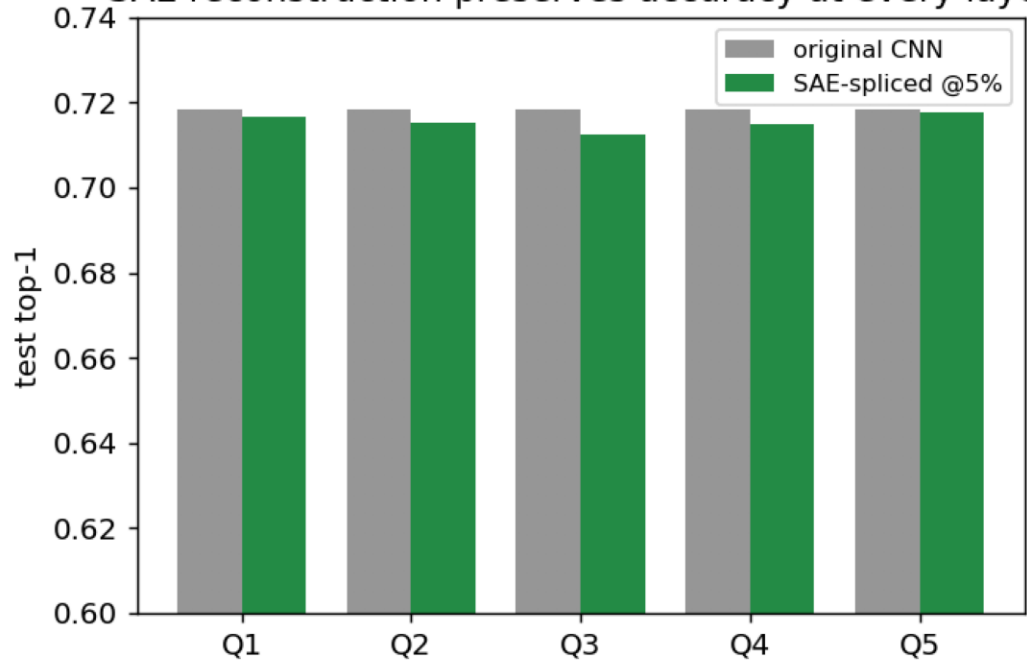
T2.2: full sparsity-fidelity Pareto (Field SAE, global mask)



Logits match the original (Q5 SAE @5%)
pred-agreement 97.6%

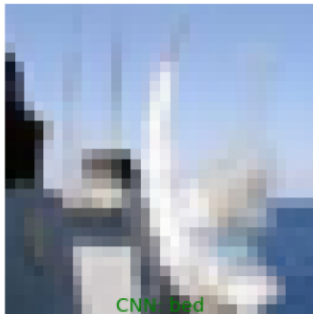


SAE reconstruction preserves accuracy at every layer



Original CNN prediction vs SAE-spliced prediction (Q5 @5% retained)

CNN: rocket
SAE: rocket



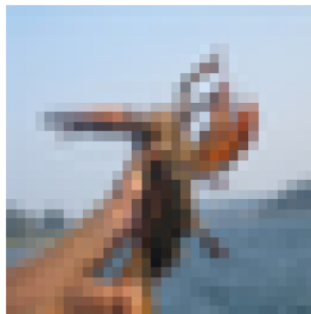
CNN: bed
SAE: bed



CNN: squirrel
SAE: squirrel



CNN: palm_tree
SAE: palm_tree



CNN: tractor
SAE: tractor



CNN: crab
SAE: crab



CNN: spider
SAE: spider



CNN: bus
SAE: bus



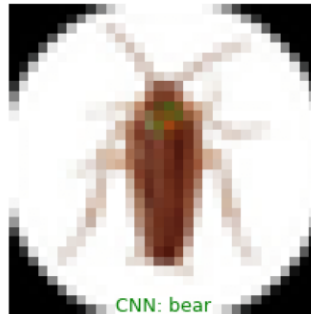
CNN: leopard
SAE: leopard



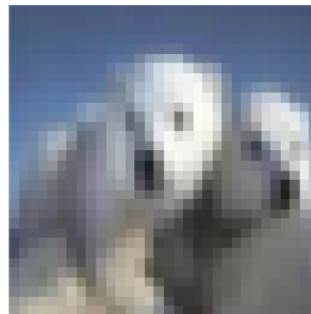
CNN: bus
SAE: bus



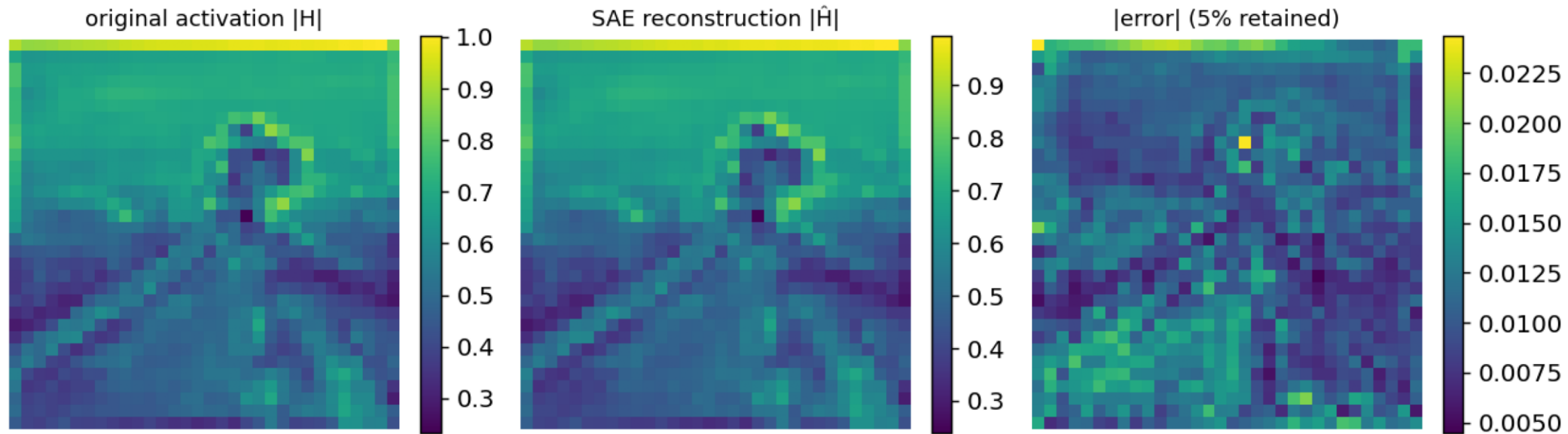
CNN: cockroach
SAE: cockroach



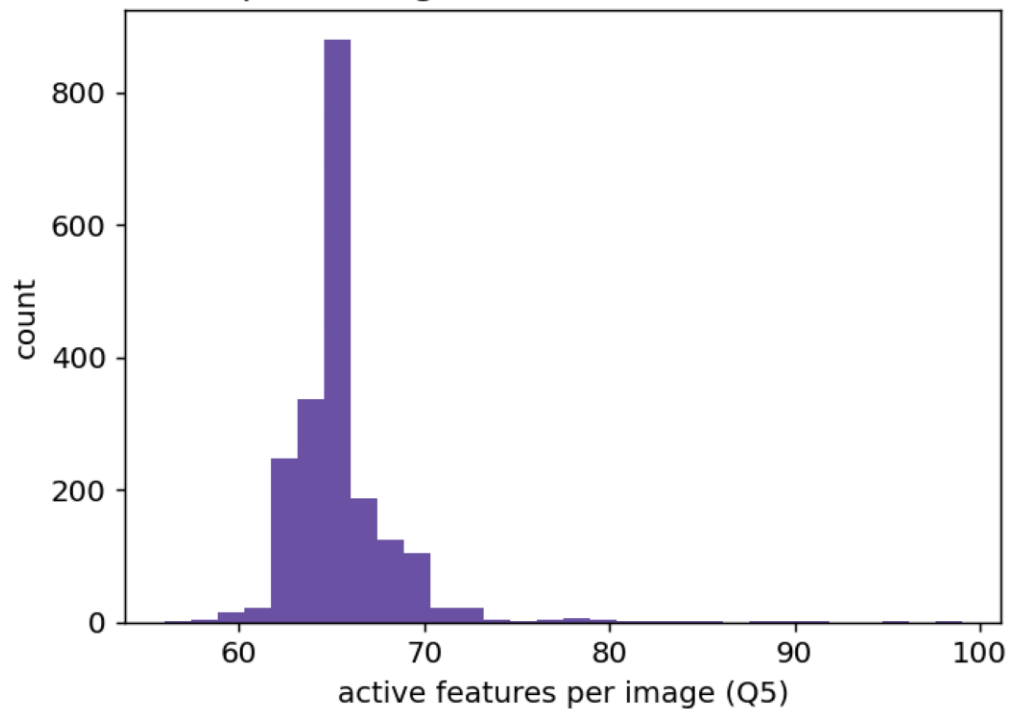
CNN: bear
SAE: bear



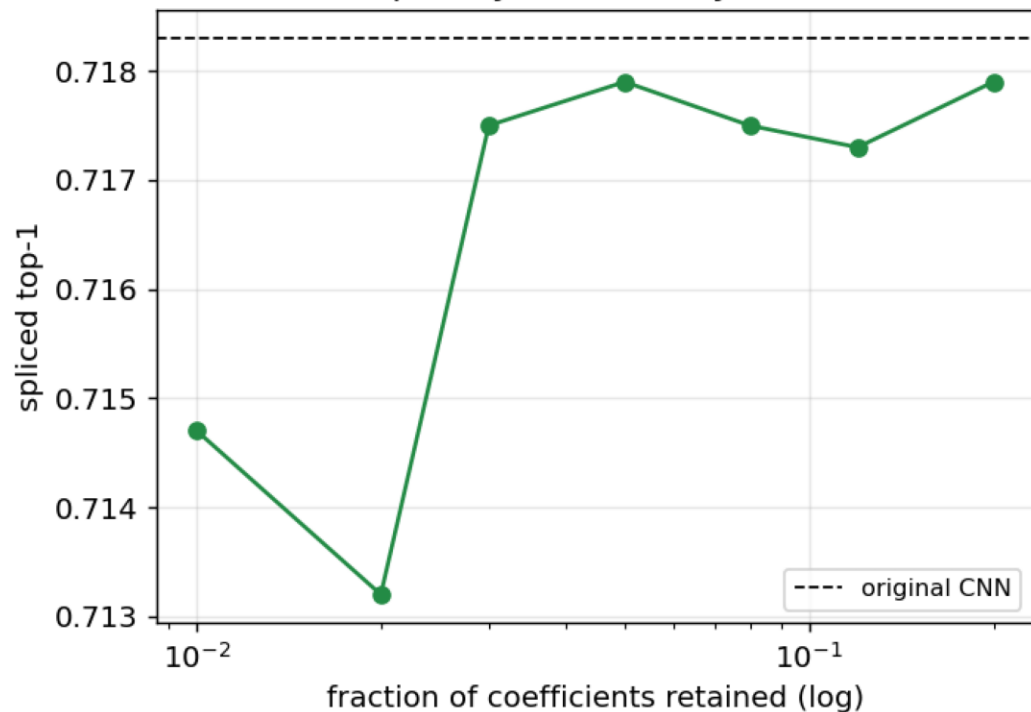
Q1 activation field: original vs SAE reconstruction (one image)



Sparse usage: ~66/512 features active



Sparsity vs accuracy (Q5)

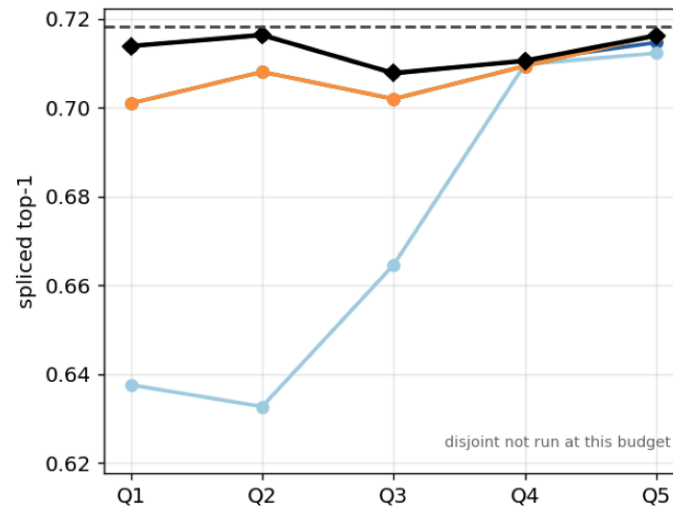


Patch/vector families by budget: zoomed against the backbone upper bound

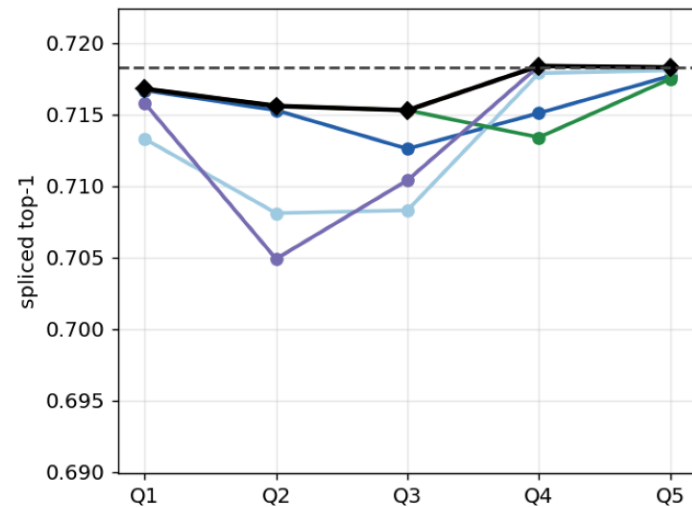
Field Overlap Disjoint
Pointwise Depthwise overlap Best observed family

Backbone upper bound

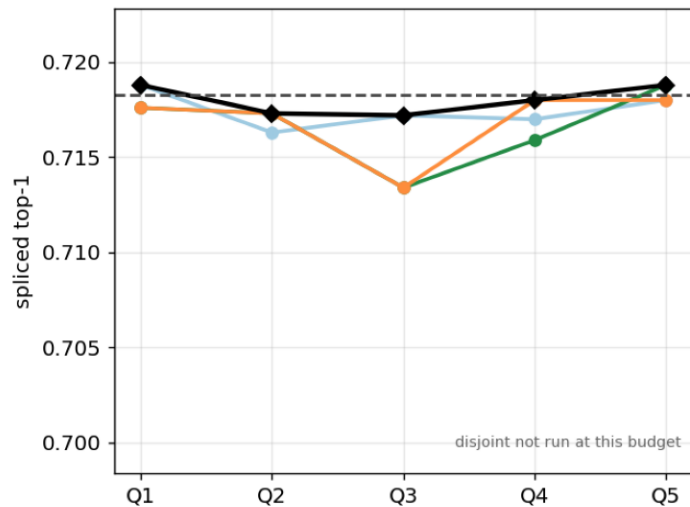
2% retained



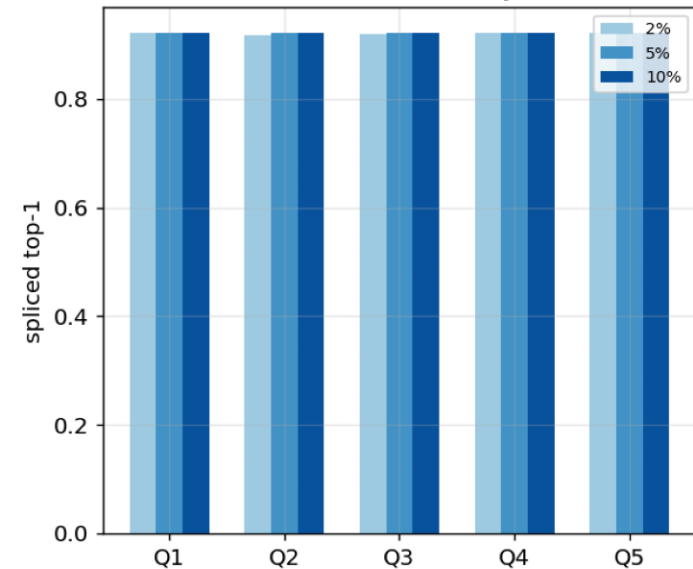
5% retained



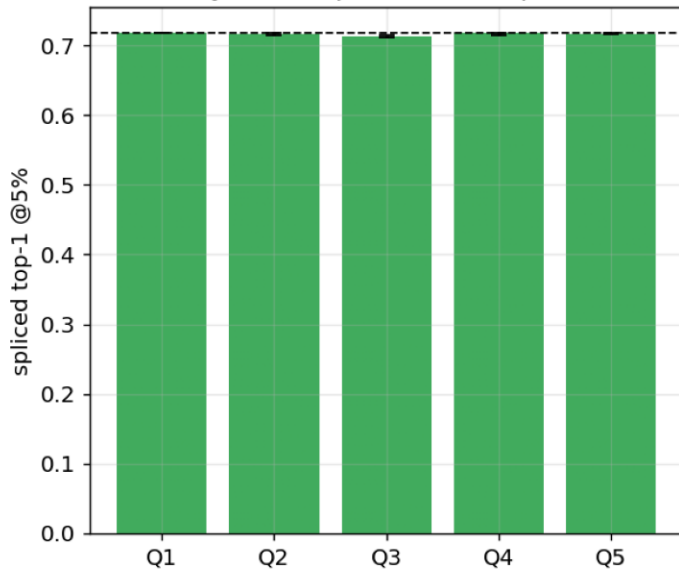
10% retained



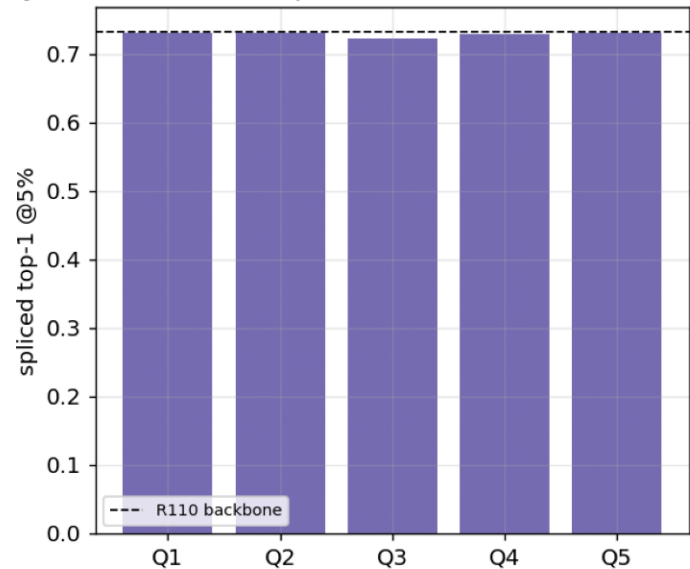
R20/C10 remains saturated by 5% retained



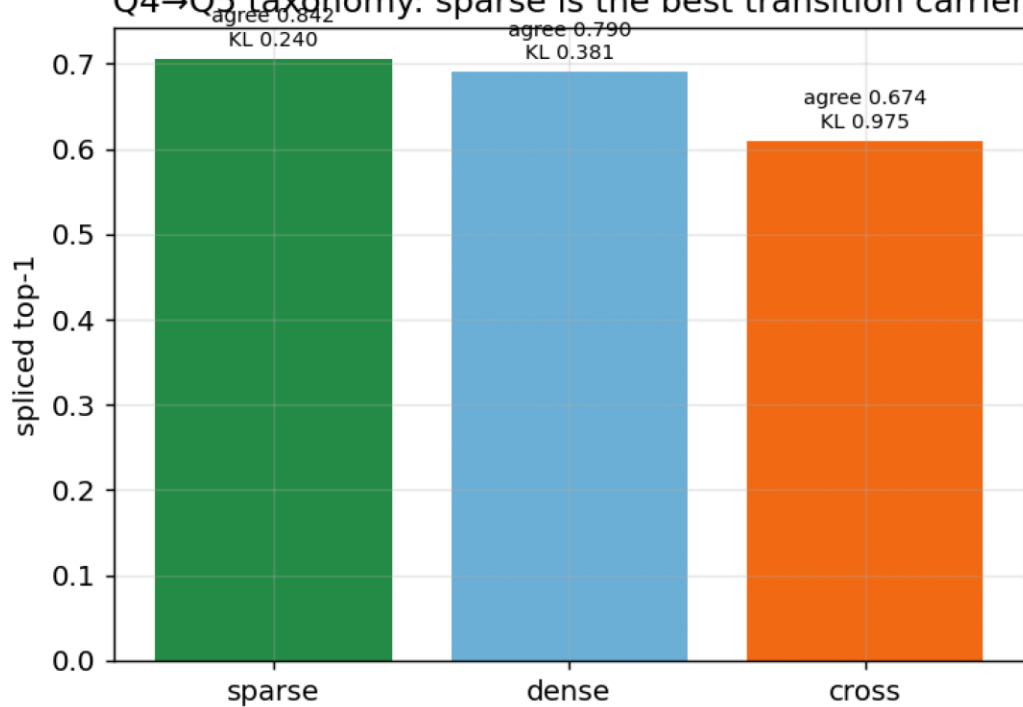
Seed stability: small spread, same qualitative story



R110 transfer keeps the same hardest section: Q



Q4→Q5 taxonomy: sparse is the best transition carrier



ViT transfer artifact works at 5% retained

